

Customer Conversion Prediction Using Machine Learning

Abstract

The objective of this project is to develop a machine learning model capable of predicting whether a customer will convert based on historical customer data. The project demonstrates data preprocessing, feature engineering, model training, evaluation, and prediction generation.

Introduction

Customer conversion prediction helps businesses improve marketing efficiency and customer engagement. This project uses machine learning techniques to analyze customer-related features and predict conversions.

Objectives

- Analyze customer data and identify relevant features.
- Handle missing values and prepare data for modeling.
- Train and evaluate classification models.
- Generate predictions for unseen customer records.
- Create a submission-ready prediction file.

Dataset Description

Training dataset contains features and target labels. Test dataset contains features without target labels. Target variable: Converted (Binary Classification).

Methodology

1. Missing Value Handling using SimpleImputer (Median Strategy).
2. Feature Selection using numerical features.
3. Feature Scaling using StandardScaler.
4. Model Development using Random Forest Classifier.
5. Training and Validation using an 80:20 split and accuracy metric.

Tools Used

Python, Pandas, NumPy, Scikit-learn, Jupyter Notebook.

Results

The pipeline successfully processed missing values, scaled features, trained a model, generated predictions, and created a submission file for evaluation.

Future Enhancements

Feature engineering, hyperparameter tuning, XGBoost, LightGBM, CatBoost, and cross-validation.

Conclusion

The project demonstrates the practical application of machine learning for customer conversion prediction and provides a foundation for future improvements.